

Phantom Clues Guide

I'm writing this as a retrospective on everything I learned while making Fog and Phantoms, both the nitty gritty of getting phantom clues to function, and some of my takeaways from my aesthetic process.

I'm going to assume you've either played the original [puzzle](#) or seen a [solve video](#). There will be minor spoilers ahead, so if you don't want the puzzle spoiled stop reading now.

Phantom Implementation TL;DR:

This section is intended for folks who are comfortable with a little JSON tinkering. I'll go over all of this in more detail later if any of it isn't clear.

Grey Phantoms (rendered above the fog)

- Grey phantom clues must be cosmetic lines.
- Their hex value should be #afafafff to match the fog.
- Their target should be cell-highlights (this will break the pen tool)

When thinking about their logic path, remember that they will be revealed when the cell is covered OR the fog is cleared from that cell.

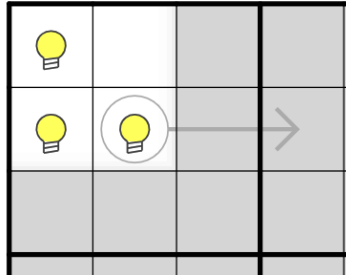
White Phantoms (rendered below the fog)

- They can be whatever type of clue, as long as it's white AND renders above the color layer
- For this reason, you may need to set them to render above the grid, which may give them away depending on how close they get to cell edges.
- They may also interact weirdly with fog, if you're using fog. I'll cover this in more depth in the aesthetics section

When thinking about their logic path, remember they will require their cell to be shaded AND the fog to be cleared before they are visible.

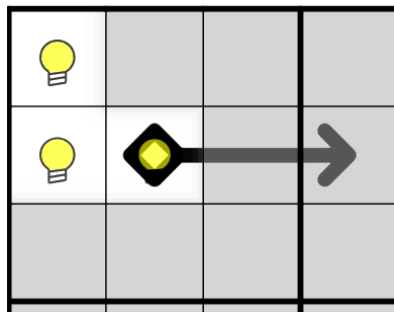
Making a grey phantom clue, step-by-step

Let's say you wanted to make this clue a grey phantom:

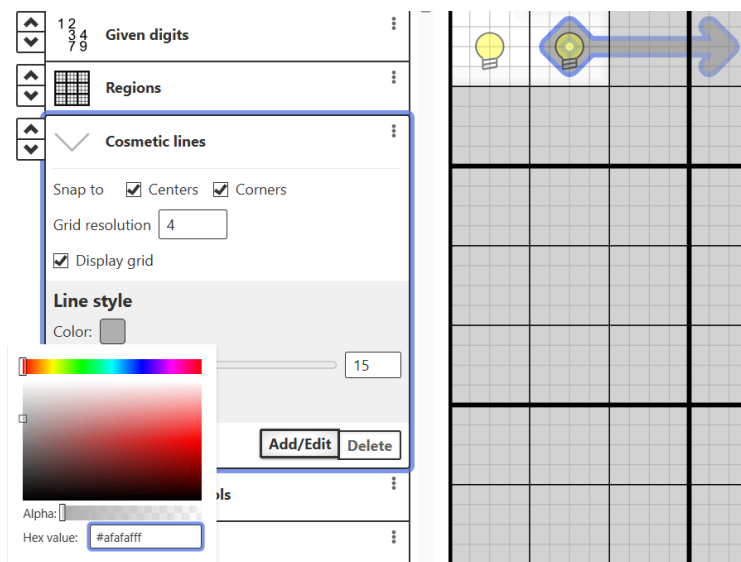


The first thing you'd need to do is draw it with cosmetic lines. This is because only lines can be assigned a target layer to render on in the JSON.

So we'll do something like this:



Don't worry about that square end, we'll fix it later. What we do want to do is make sure it's fog-grey. So we'll set the color to #afafafff

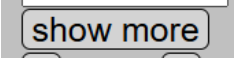


That's all I would do with it in Sudokumaker. For the next step I would use Phil's excellent [line editing tool](#).

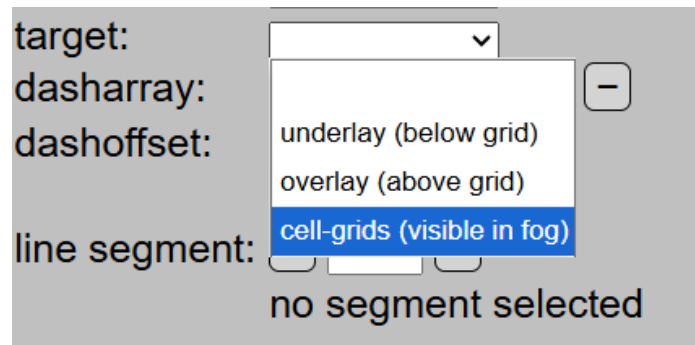
This next bit is kind of a one way street, I think, so I'd have the rest of the puzzle fully set before fully phantomizing my grey cells. When ready, we'd make a sudokupad link for our puzzle, and



then load it into Phil's tool by clicking the button in the upper left, and then providing the URL. I would probably uncheck the auto-curve lines as well, in order to keep any straight lines intact for now.

With our puzzle loaded in, click the future phantom clue, and then click  over on the left.

What we're looking for here is the target selection menu. For now, we'll use Cell-grids, which is what you would normally want to do in order to get a clue to render above the fog. This is the layer where cell and region borders are rendered:



Unfortunately, it won't work for our purposes, as [this](#) non-puzzle demonstrates.

Instead, we'll need to set the target to cell-highlights manually in the JSON. To do this, click the



in the top left, and look for the section titled 'lines'.

```
"lines": [{"color": "#afafafff", "thickness": 9.6, "d": "M208 80L224 96L208 112", "t": "208 80 224 96 208 112"}, {"color": "#afafafff", "thickness": 9.6, "d": "M224 96L128 96Q112 96 96 112L80 96L96 96", "t": "224 96 128 96 112 96 96 112 80 96 96"}, {"overlays": [{"center": [1.5, 5.5], "width": 0.82, "height": 0.82, "thickness": 1.28, "color": "#afafafff"}]}
```

We'll be replacing the selected word 'grids' with 'highlights'

You may notice there's a second line object. That's because our arrow is rendered as two separate lines even though it displays as a single object in sudokumaker. You'll need to set both targets to cell-highlights. You will also find that the "target": field doesn't exist in the JSON if you haven't changed it in any way. That field only populates if you change it from default, so I always change phantom clues to cell-grids initially so that I know where I'm aiming for my cell-highlights changes.

NOTE: Giving any clue a cell-highlights target will break the pen tool because you're putting clues in places Sven wasn't expecting them.

And that's it! I'll go over making the arrow's head a circle later on in the aesthetics portion of the

guide, but in terms of getting the clue in the right layer, you're all done. You can click the button to generate a new sudokupad link with your changes intact.

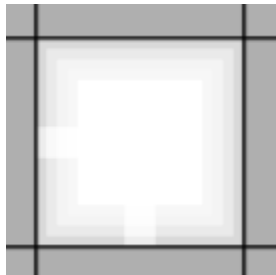


White Phantoms and Fog

Keen eyed solvers may have noticed something funny with the white clues in Raising Sails



If you were wondering why they faded as they approach the edge of the cell, let me show you what the unshaded cell would look like without that fading:



Not very stealthy, if you want that information to be concealed until the cell is shaded. The only way I was able to work around it was to create a bunch of overlapping line clues, with increasing transparency as they approached the edge. All told, I wound up using 48 lines for that one arrow, which was definitely overkill, and could almost certainly be automated by a superior programmer than me. Anyone who has a better method, please let me know.

The end result isn't perfect.

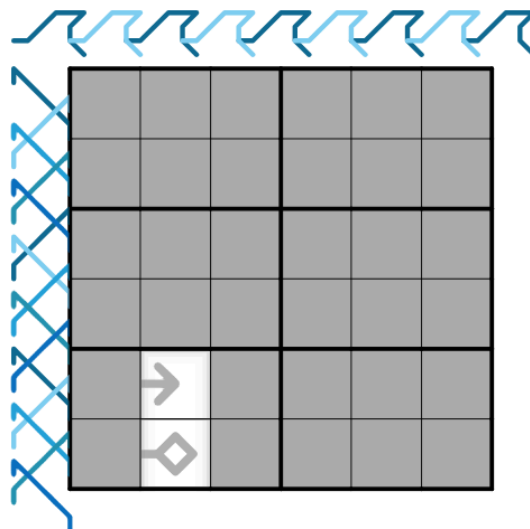


If you look very closely, you can still see the edge of the line going south, but I think you have to be working pretty hard to cheat this way, and that's good enough for me.

Luckily this is only necessary for white clues that you expect to be next to a foggy. In the other puzzles, I also use white clues, but the solve path required the relevant cells to already be shaded by the time the cell was revealed, so this labor was only needed once.

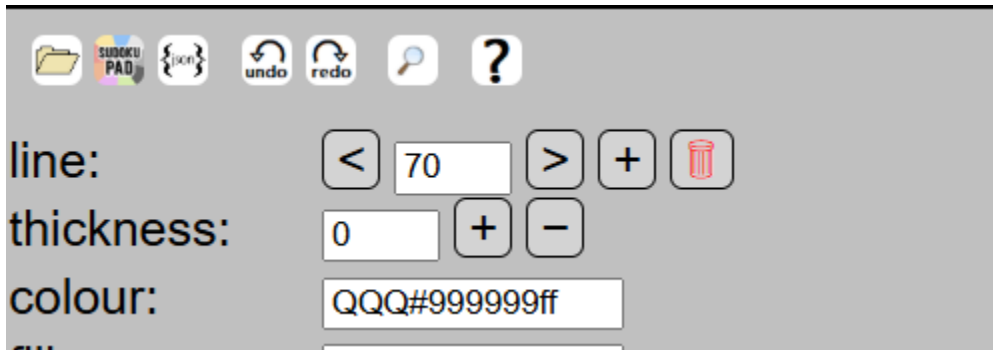
Phil's Tool, Tips and Tricks

This section is just intended as a collection of techniques I picked up over the course of the project that improved my workflow. Many of them will be simple or obvious, but I'm passing them along in case they're useful. 90% of the aesthetic work was done in Phil's tool. The final sudokumaker project for Raising Sails looks like this:



The first thing that made my life easier was recognizing that the JSON editor allowed me to copy lines in place. Simply finding the relevant line in the JSON, copying it, and then pasting it in place creates a duplicate in the editor that can then be moved and edited. This meant I didn't have to have all my objects pre-populated before starting the aesthetic pass.

One tricky part was identifying the specific line I was looking for in the JSON. I found it helpful to select the line I wanted and then change its color to something easy to hunt for in the JSON:



Placing the QQQ before the color makes it stand out among all its brothers:

```
{ "color": "#b9698c99", "target":  
  { "color": "#999999ff", "thickne:  
    { "color": "QQQ#999999ff", "thicl  
      { "color": "#999999ff", "thickne:  
        { "color": "#116d9399", "thickne:  
          { "color": "#116d9399", "thickne:  
            { "color": "#116d9399", "target":
```

When I'm done copying and pasting it, I can get rid of the extra Qs right there in the JSON. Mostly. If you hunt around inside the final puzzles, you'll definitely find places where I forgot...

This same technique can be used for organization, if your line numbers are getting north of 200. This isn't something I thought to do at the time, but would have been a huge help if I had. As you can see, the thickness of the line is 0, which means the line itself never renders. All that matters is its fill, which means the color can be named anything and have no effect. In the future I expect I'll give the topmost cloud layer a color named something like 'clouds →' and the bottommost '← clouds'. Obviously, this only works if you aren't using the line as a line, but you could always add dummy lines with zero thickness whose only purpose is to keep your line hierarchy clear.

My other aesthetic thoughts are more impressions than techniques.

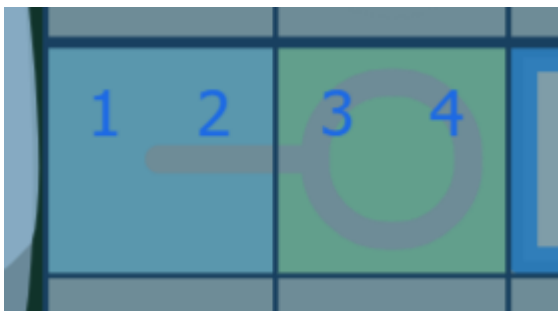
I used the cell grids layer to make my border art because I liked the way it broke the 4th wall when it overlapped the grid.



←Fun sense of depth?

I felt like the white grid ruining the atmosphere I was working to creat when I was making my cave puzzle, so I wound up making a color wash by fitting a square over the face of the grid in the cell-grids layer with a relatively high transparency, and then used the same effect in the other puzzles to make them mast and moonlight flavored. This, coupled with the fog, makes shading tricky, so if you do something similar I recommend making a version without the washes for anyone who might need it (I have no idea if anyone took me up on my art-free versions of the puzzle, but I definitely rested better having made them).

In general, a lot of native shading options show up poorly for grey phantom clues because the shades are in the same range as the grey fog. I found creating a palette of darker shades in sudokupad really helped with visibility (and for once, enabling outlines on elements helped my eyes).



vs.



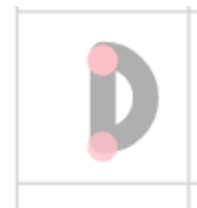
Oh right, and my current technique for making an aesthetic lines circle:

Firstly, don't draw a square, draw a line to act as the diameter. This will mean you only have to deal with making 2 semi circles, rather than 4 quarter circles. In Phil's tool select the line and navigate to the second line segment:

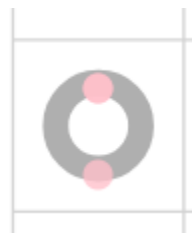
line segment: < 2 > +
type: line
coord type: absolute
x: 32
y: 112

Hit the plus next to the line segment, which will add a third line segment. Change that segment's type to arc, and then set that segment's x and y coordinates to match those of the first line segment (you now have a line that crosses the circle's diameter and then arcs back to its starting point). You'll need to set rx and ry to both be the circle's radius.

line segment: < 3 > + 
type: arc
coord type: absolute
rx: 16
ry: 16
rot: 0
lge arc flag: <180°
sweep flag: negative
x: 32
y: 80



Then cycle back to segment 2, change its type to arc as well, and match your radius. Voila, cosmetic line circle.



I think that's all I have for you. I'll finish by sharing the various in-progress sudokupad links for anyone who wants to see how the artwork came together. If you have any questions or corrections let me know, and I'll happily update and elaborate. Cheers, thanks for read, and happy setting!

Dubious

Raising sails:

<https://sudokupad.app/rz7wkv00fo>

<https://sudokupad.app/cf9undsf5y>

<https://sudokupad.app/tavgruduht>

<https://sudokupad.app/299hvf606a>

<https://sudokupad.app/oukul045vg> (adding the fading white arrow)

<https://sudokupad.app/xqs9sbwudh>

Tunnel Vision:

<https://sudokupad.app/s5i0cp0ud4>

<https://sudokupad.app/n36ss4w0tr>

<https://sudokupad.app/l0rbdwhb3k>

<https://sudokupad.app/azn95tkpe1>

<https://sudokupad.app/ep6vngod8i>

<https://sudokupad.app/h3vy0404wq>

<https://sudokupad.app/zfwlb2n3a0>

<https://sudokupad.app/s5jazqs1y2>

<https://sudokupad.app/4nj8vkt1ze>

<https://sudokupad.app/ywbmdcj8re>

Temple Ruins:

<https://sudokupad.app/liowhii9fm>

<https://sudokupad.app/q24xfskpk6>

<https://sudokupad.app/u8x4h7wwqp>

<https://sudokupad.app/4lqpn3qb34>

<https://sudokupad.app/c6zv0k4wur>

<https://sudokupad.app/zr5o4fjigw>

<https://sudokupad.app/5klr77nti6>

<https://sudokupad.app/j5eq271wnw>

<https://sudokupad.app/4ehlnci1jq>

<https://sudokupad.app/qpu5re1cfp>

<https://sudokupad.app/w8umas41m3>